

Nikita Vostrosablin, PhD

Engineering Leader focused on production ML and backend systems. I bring experience from both research-heavy and product-focused environments.

PRAGUE, CZECHIA

CONTACT

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EXPERIENCE

Stats Perform

> Engineering Manager Jul 2025 - Present | Prague, Czechia

Tech: Python (FastAPI, Pydantic, scikit-learn, pandas, PyTorch), Bash, SQL, PostgreSQL, AWS, Kubernetes, Kafka, Spark, Terraform

Skills: Team management, Project management, Stakeholder management, Incident management, System design

- Lead **9-12** engineers owning architecture, reliability, and operational health for **~30** production **ML/data** pipelines and **~150** distributed Python services across **AWS** and **on-prem Kubernetes**.
- Drove architectural decisions for new and existing production pipelines at **multi-team** scale, partnering with **Product**, **Data Engineering**, and **AI** teams to align stakeholders and technical direction.
- Defined SLOs, observability, alerting, and on-call practices, cutting incident resolution from **multiple** days to **~1** day and support load from **~25%** to **~2%** of team capacity.
- Improved key pipeline processing times by up to **~80%** through architectural redesign and algorithmic optimization.
- Hired **3** engineers, led **3** promotions, introduced squad-based ownership and the **DX4** framework, and drove AI-assisted workflows where **~75-80%** of code is generated by agents under controlled review.

> Senior Software Engineer, Team Lead Dec 2023 - Jul 2025 | Prague, Czechia

Tech: Python (FastAPI, Pydantic, scikit-learn, pandas, PyTorch), Bash, SQL, PostgreSQL, AWS, Kubernetes, Kafka, Terraform

Skills: Microservice Architecture, People leadership, Technical leadership

- Acted as technical lead for a **4-engineer** sub-team productionizing **ML** pipelines and distributed systems for [Opta Vision](#), a **widely used sports analytics product**.
- Designed and launched **event-driven ML pipelines** and services for **real-time** and post-game metric generation, rebuilding underperforming flows to improve throughput.
- **Led migration** of compute workloads from AWS-managed services to Kubernetes as part of a broader cost-optimization effort.
- **Drove architecture decisions**, introduced design-first development through design docs, mentored engineers, and contributed to hiring (*example of impact*: reduced cost of major pipelines by **85%**).

Merck (MSD Czech Republic)

> Scientific Software Engineer, Team Lead - Jul 2021 - Nov 2023 | Prague, Czechia

Tech: Python (FastAPI, Pydantic, scikit-learn, pandas, PyTorch), AWS (Lambda, Step Functions, Batch Jobs), Celery, Redis, Apache Airflow, Nextflow, Terraform

Skills: Project Management, Technical leadership

- Led **2 contractor software engineers** delivering scientific software and platform capabilities for drug discovery.
- **Launched a platform** for mRNA molecule optimization and participated in **open-sourcing** part of it as [mRNAid](#), an experimentally validated open-source tool.
- Contributed to **patented** systems for drug discovery (Reaction Synthesis Pathfinder, Publication No. 20240105285; CMC Change Assessment System, Publication No. 20240004935).
- Built AWS event-driven genomics platforms with Lambda, Step Functions, Batch Jobs, Celery, Redis, Apache Airflow, and Nextflow.
- Introduced Terraform, CI/CD, and production observability for scientific platforms and genomics pipelines.

> Python Developer / Machine Learning Engineer May 2019 - Jul 2021 | Prague, Czechia

Tech: Python (scikit-learn, pandas, PyTorch, TensorFlow), Jenkins, MLflow, Databricks

Skills: Machine Learning, Data Science

- Worked across **AI** and bioinformatics departments on molecular computation and optimization problems for drug discovery.
- Participated in **open-sourcing** [Mutation Maker](#), an application for mutagenic primer design used in protein engineering workflows.

Dimension Data

> Data ETL Engineer Dec 2017 - Apr 2019 | Prague, Czechia

Tech: Python, Apache Airflow, Jenkins, PowerBI, OracleDB

- Built **BI infrastructure** for a **major European client**, covering automated dataflows, reporting, and data-quality processes.

ORIFLAME SOFTWARE

> Data Scientist Jun 2017 - Aug 2017 | Olomouc, Czechia

Tech: Python (scikit-learn, pandas, NumPy), NLP, Machine Learning

- Applied **ML** and **NLP** methods to **sales forecasting** and review sentiment analysis, later integrated into worldwide sales operations.

Freelance

> Freelance Web Developer Sep 2016 - Dec 2016

Tech: Node.js, HTML, CSS preprocessors, Template engines

- Built web tooling and automated mail reporting for an appartement renovation web service.

EDUCATION / ACADEMIC WORK

- **Univerzita Palackeho v Olomouci**
PhD, Quantum Physics (Sep 2014 - Mar 2021)
- **Lomonosov Moscow State University (MSU)**
Master of Science, Physics (2008 - 2014)
- **DTU - Technical University of Denmark**
Researcher, quantum optomechanics (Sep 2016 - Nov 2016)

LANGUAGES

- Russian (Native)
- English (Fluent, Professional Working)
- French (Limited Working)
- Czech (Intermediate)

TECHNOLOGIES

- **Languages & Core:** Python, Go, SQL, Bash, C++, JavaScript
- **Cloud & Infrastructure:** AWS, Kubernetes, IaaS (Terraform), CI/CD (Jenkins, ArgoCD)
- **Streaming & Messaging:** Kafka, Apache Flink, SQS, SNS
- **Databases:** Relational (Redshift, PostgreSQL, MySQL), NoSQL (DynamoDB), Graph (Neo4j)
- **ML / Data Science:** pandas, Polars, scikit-learn, PyTorch
- **Orchestration:** Airflow, Temporal
- **Observability:** ELK, Grafana, OpenTelemetry

CURRENT HOBBIES

- Writing a small compiler in **Go**.
- Creating small games in the **Godot** engine.
- Composing music.
- Rating movies.